When Benchmark Inferences Do Not Compose: Projectibility in AI Evaluation

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Data, materials, and code availability. The empirical companion, including source versions, data hashes, and estimator code, is publicly available at https://github.com/BrettRey/benchmark-inference-composition. The reanalysed trial data are the public release accompanying Zhang, Koyejo, and Yang (2026). No new human-subject or proprietary data were generated.

Author contributions. Sole author. Brett Reynolds conceived the framework, developed the formal apparatus, constructed the worked example, wrote the estimator code and known-truth demonstration, and drafted and revised the manuscript.

Use of large language models. The large language models ChatGPT 5; Claude Sonnet 4.5 and Opus 4.8; Gemini 3.1 Pro served as drafting and editing aids throughout the preparation of this paper. Their use went beyond AI-assisted copy editing and is therefore declared here in full. The author is responsible for all theoretical claims, arguments, errors, and interpretive choices, and affirms that the final text reflects his own work. No language model is listed as an author, and none satisfies the journal's authorship criteria.

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